

Hardware Reference Manual

Redstone

VOLTAGES **3**

Input Power Requirements 3
Auxiliary Voltage Output 3

BACK PANEL LAYOUT **4**

CLUSTER CABLING **5**

System Cluster 5
Transmitter Cluster 6
Receiver Cluster 7
Acquisition Cluster 8
Gradient Cluster 9

NEW FEATURES WITH EM-III **10**

Extended Tables 10
Conditional Branching 12
Simultaneous Tables 16

MODULES **17**

Digital Attenuator 17
Utility 18

LapNMR

VOLTAGES **19**

Input Power Requirements 19
2U Chassis Auxiliary Voltage Output 19
Portable Chassis Auxiliary Voltage Output 19

BACK PANEL LAYOUT—2U CHASSIS **20**

BACK PANEL LAYOUT—PORTABLE **21**

SPECIFICATIONS **22**

Pulse Programmer 22
Data Acquisition 22
Transmitter (Analog) 23
Transmitter (Digital) 23
Digital Receiver 24
RF Receiver 24
Power Supply 25

CONTACTING TECMAG **26**

Redstone

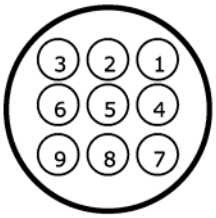
Voltages

Input Power Requirements

Voltage: 100 – 240 V_{AC} auto-switching
Frequency: 50 – 60 Hz
Power: Nominally 600 W depending on configuration

Auxiliary Voltage Output

AUX 1&2



1. +5V
2. 5V GND
3. -5V
4. +15V
5. 15V GND
6. -15V
7-9. N\C

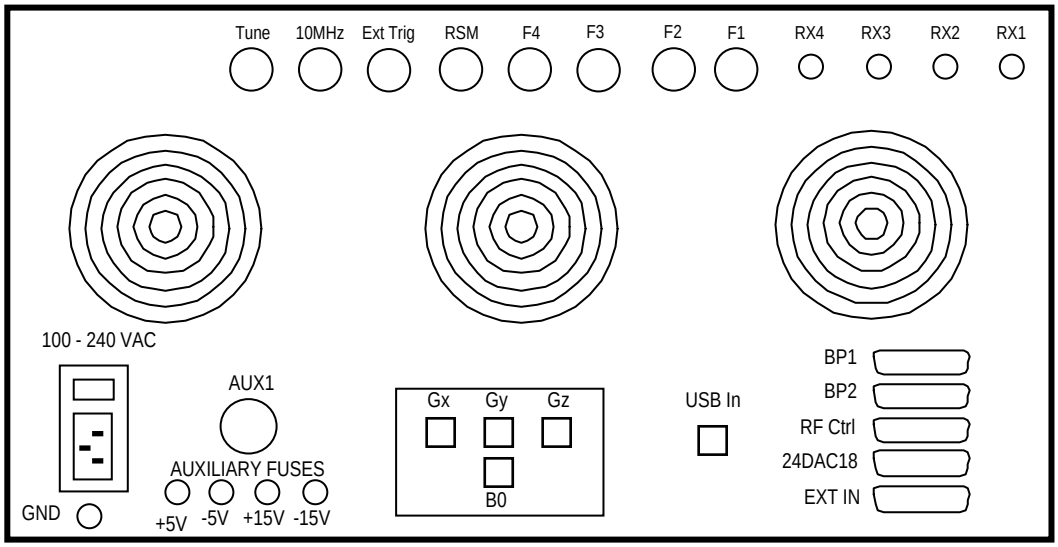
Auxiliary Current:

Voltage	Max. Current (A)
+5V	3
-5V	0.5
+15V	0.5
-15V	0.5



Rear panel Redstone Auxiliary DC Amphenol power connector & fuses

Back Panel Layout*

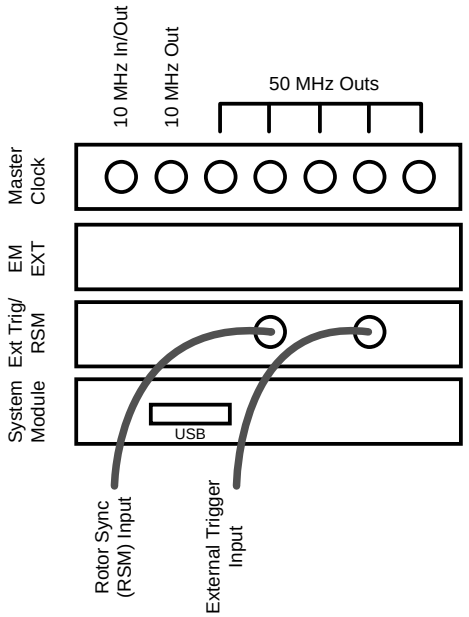


RX1 – RX4:	Receiver inputs
F1 – F4:	Transmitter outputs
RSM:	Rotor Synchronization Module input from probe tachometer circuitry or Tecmag Air Control
Ext. Trig:	External Trigger input; TTL; for external triggering of the Redstone pulse programmer
10 MHz:	10 MHz clock output (internal clock configuration--default) 10 MHz clock input (external clock configuration)
Tune:	Optional RF output to Tecmag Probe Interface or directional coupler for RF tuning
BP1 and BP2:	Pulse programmer-controlled TTL outputs for amplifier blanking and scope triggering; additional lines may be added as required
RF Ctrl:	Optional latched voltage output for Tecmag Probe Interface
24DAC18:	Optional 24 channel DAC output for Tecmag SU-24 room temperature shim controller
EXT IN:	The EXT IN 25-pin connector provides inputs for conditional branching hardware tests. Pins 1-4 are bits 0-3, respectively; pin 25 is a ground (GND).
G_X, G_Y, G_Z, B_0:	Digital (RJ-45) or Analog (Twin-axial BNC) gradient signal outputs
AUX1 and fuses:	0.5 amp +5, -5, +15, and -15V_{DC} outputs for auxiliary devices including preamplifiers, Tecmag Probe Interface, and Tecmag Air Control

* Exact connector location & labeling subject to change. Consult installation documentation.

Cluster Cabling

System Cluster



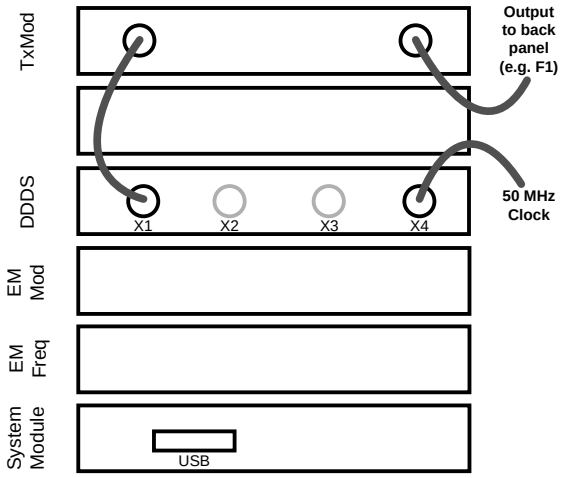
(Optional) **EM EXT** provides latched voltage output lines for Tecmag Probe Interface, RF relay control, etc.

Master clock: Default configuration is internal clock with the first 10 MHz port defined as an output. It is possible to switch to external clock mode (contact technical support) in which case the first port becomes a 10 MHz input for synchronizing the Redstone to another device.

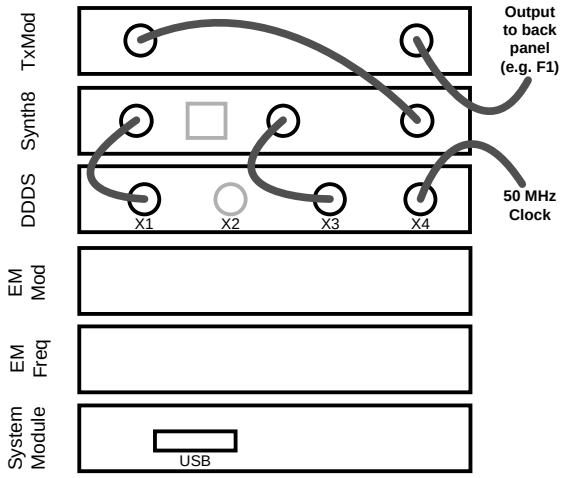
Ext Trig/RSM (shown here) or Utility: In 2012, the Ext Trig / RSM module was replaced with a Utility module. A ribbon cable header is used for inputs rather than SMB connectors

Transmitter Cluster

Low Frequency



High Frequency



Low Frequency: 0.1 to 125 MHz[†]
High Frequency: 5 to 500 MHz[*]
Max Output: At least 0-4 dBm across frequency range
Max Rise Time: <30 ns @ 100 MHz

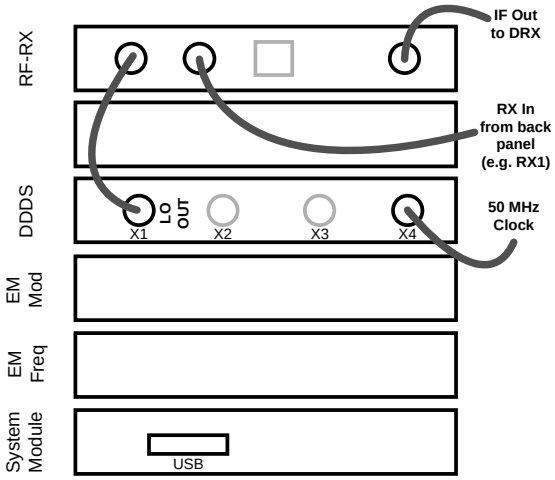


A high frequency Transmitter cluster

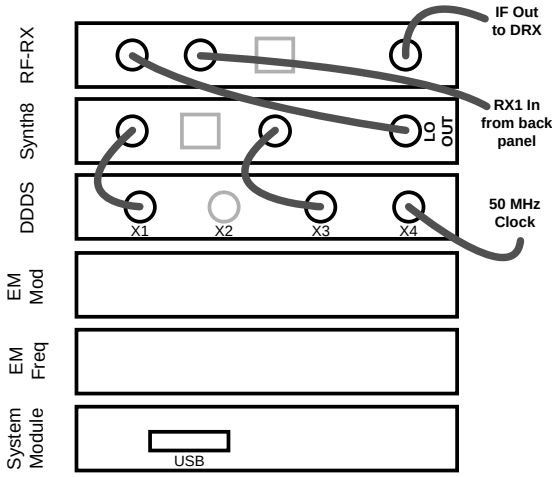
[†] Extended frequency ranges available upon request

Receiver Cluster

Low Frequency



High Frequency



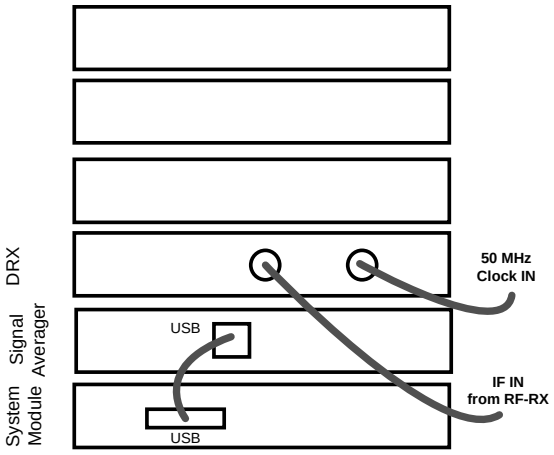
Single receiver configurations are shown here. When configured for multiple receivers the RF-RX analog receivers are located in the Acquisition cluster rather than the Receiver synthesizer cluster (See Acquisition Cluster—Dual Receivers below).



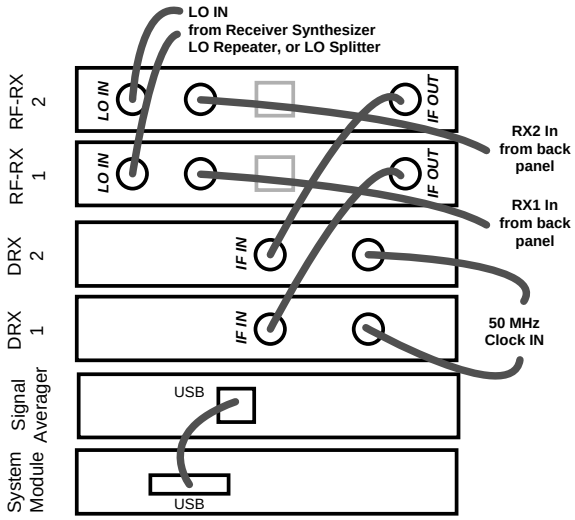
A high frequency Receiver Cluster for a single receiver Redstone

Acquisition Cluster

Single Receiver



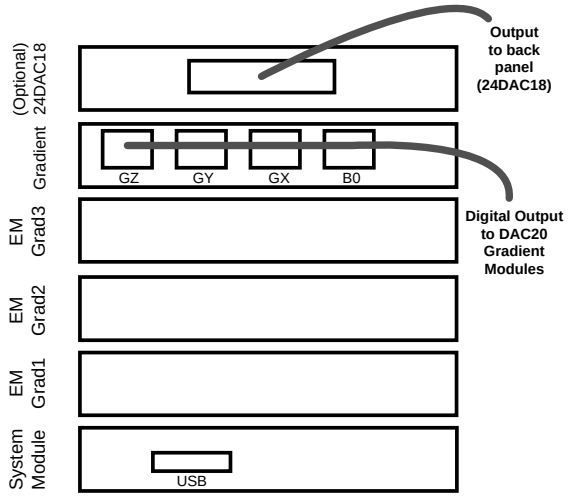
Dual Receiver





An Acquisition Cluster for a single receiver Redstone

Gradient Cluster



The digital output of the Gradient cluster is delivered to DAC20 modules (not shown) using RJ-45 cable that has been modified at the factory to inhibit ground loops. Depending on the configuration the DAC20 modules may be located in a separate 2U rack-mountable DAC chassis, mounted within the Redstone chassis, or installed within certain gradient amplifier models.



A Gradient Cluster with optional 24DAC18

New Features with EM-III

The Tecmag Redstone system is based on the latest Event Module—the EM-III. Below are detailed a few of the new features made possible with the EM-III including *Extended Tables, Conditional Branching,* and *Simultaneous Tables.*

Extended Tables

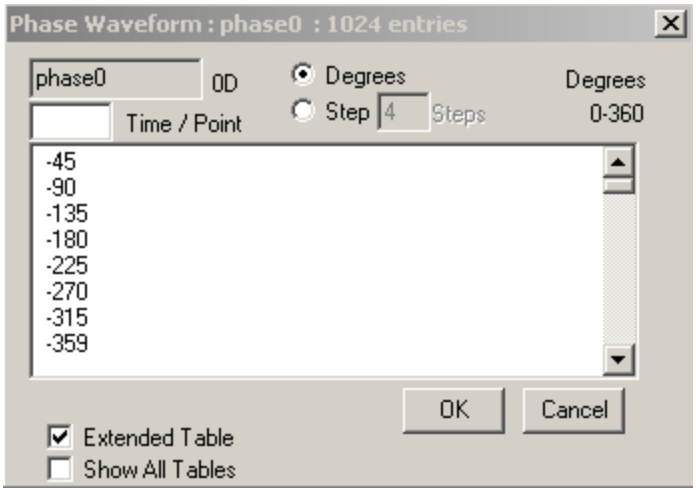
Certain table types may now be "extended". This means that the table pointer is no longer reset along with the associated scan counter. Thus, (depending upon the number of entries) the table entries can become "asynchronous" from the scan counter. For example, suppose it is desired to apply a pseudo-random phase to the transmitter and receiver on each excitation, to accomplish so-called "rf-spoiling" of the transverse magnetization. In such a case, it would be undesirable for this phase pattern to repeat itself synchronously with the phase encoding in a 2D experiment (e.g., every 4th record). If we use an extended table, we can accomplish this, as illustrated in the chart below, which compares the phase from the following tables:

- Standard 1D table (0º, 180º)
- Standard 2D table (0º, 90º, 180º, 270º)

  vs.
- Extended 1D table (117º, 234º, 351º, 108º, 225º, 342º, 99º, 216º, 333º, 90º, 207º, 324º, 81º)

1D scan #	2D scan #	Phase from standard 1D table	Phase from standard 2D table	Net phase from standard tables	Phase from 1D extended table
1	1	0	0	**0**	**117**
2	1	180	0	**180**	**234**
1	2	0	90	**90**	**351**
2	2	180	90	**270**	**108**
1	3	0	180	**180**	**225**
2	3	180	180	**0**	**342**
1	4	0	270	**270**	**99**
2	4	180	270	**90**	**216**
1	5	0	0	**0**	**333**
2	5	180	0	**180**	**90**
1	6	0	90	**90**	**207**
2	6	180	90	**270**	**324**
1	7	0	180	**180**	**81**
2	7	180	180	**0**	**117**
1	8	0	270	**270**	**234**
2	8	180	270	**90**	**351**

In this example, 2D records 1, 5, and so on have the same phase with the standard tables. The 1D extended table repeats on the second scan of the 7th record, resulting in far less correlation with record number. Note that because this table is a **1D extended table**, the pointer increments with the **1D scan counter**.



To specify extended table behavior, check the box in the table editor, as shown above. This feature is available for "Icon types" A3, E3, G3, P3, R3, and W3.

Other notes:

- All tables in the same dimension, on the same event, and on the same module (e.g. amplitude and phase), must have the same setting of the "extended" flag. The compiler creates a composite table for this event, and uses the flag value from the last table it processes.
- The number of table entries should be different from the corresponding Scans nD or Points nD, in order to be useful.

Conditional Branching

Conditional branching allows the execution of events in a pulse sequence in a different order, depending upon some **condition**. Such conditions might include:

- Scan count values
- Presence or absence of an external (hardware) signal (future)

The changed order of events results from a **branch**, from a branch event to a target event (The target is normally some event other than the next one in the sequence).

To allow the increased flexibility in pulse programming that results from conditional branches, some hardware and software features have been added to the Redstone system:

- A hardware state or condition input distributed to every EM-III (implemented with PCB rev C)
- A **CTest** sequence line to specify when the condition input is to be tested or detected, and what sort of test to perform. A CT event specified on this line will set an internal flag bit = 1 if the condition is true. The flag state is stored and persists until the next CT event occurs.
- A **CBranch** sequence line to specify the branch destinations. Tables on this line indicate the target event(s) for a branch. A branch occurs if the flag (from CTest) is true when the branch event is executed.

Some rules:

- Conditional branches apply only to Redstone/EM-III systems.
- Minimum time for CB/CT events is 1μs
- The condition must be tested BEFORE the branch event.
- The branch event and its target are always executed.
- Branches are always specified in tables. All branches are specified by the target event number as shown in the sequence editor. (If the number of events in the sequence is changed, the sequence editor will attempt to update the target event numbers, and warn the user if this is not possible).
- No branching INTO or OUT OF a loop (an entire loop may be skipped, however).
- No branching into another scan: i.e. beyond the START or END-OF-SCAN events for the current scan.
- No branch can skip any slow-io event (frequency hop, gradient rotation, or acquisition).
- nD tables: the highest dimension overrides any lower dimension branches (tables replace rather than add).
- A branch may not have a waveform table on the same event. However, the branch's target may have a waveform.
- A branch may not occur in the same event as an external trigger event.
- There is no protection against infinite loops

Limitations:

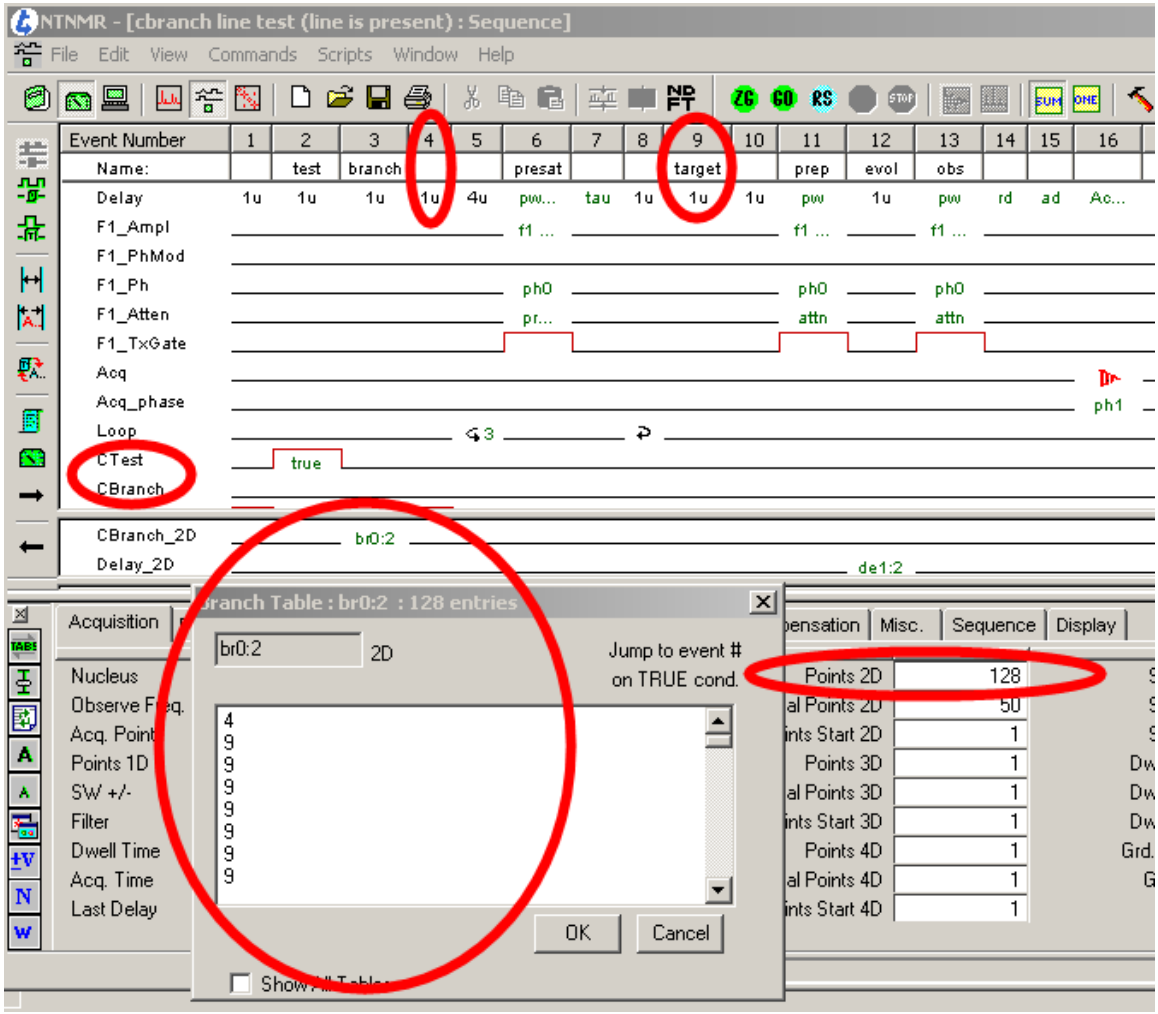
There are presently no software protections against illegal branches, infinite loops, or other problems that might cause the pulse programmer to lock up. Users must test their sequence under safe conditions (e.g. with the power amplifiers disabled).

Example 1: 2D Preparation Sequence

The following example performs the presaturation loop (events 5-7) only once for each 2D plane. On the first scan, events 4-8 are executed (the branch is from event 3 to 4). On all subsequent scans, a branch from event 3 to 9 causes events 4-8 to be skipped. Note the extra event (#4) inserted before the loop start to provide a target for the first jump.

Pulse Programming procedure:

In event 2, place "Always True" on the CTest line.
In event 3, place a 2D table on the CBranch line.
Put the following entries in the table: 4, 9, 9, 9…. (the target event numbers, total 128 entries)



Example 2: Conditional (“Gated”) Acquisition

The following example shows how to maintain a steady-state magnetization while acquiring signals only when an external hardware condition is satisfied. This is useful, for example, in respiratory gating, where data are only acquired during a restricted phase of the subject’s breathing cycle.

In the example, events 8-13 and 14-19 have exactly the same time duration. Depending upon the state of the external hardware, the sequence will either:

- execute events 8-13, then branch to event 1 (branch2 – target2, in red), without incrementing the counters, and then repeat the same scan, or
- branch from 8 to 14 (branch1 – target1, in blue), then execute events 14-19, then increment the 1D/nD counters and advance to the next scan.

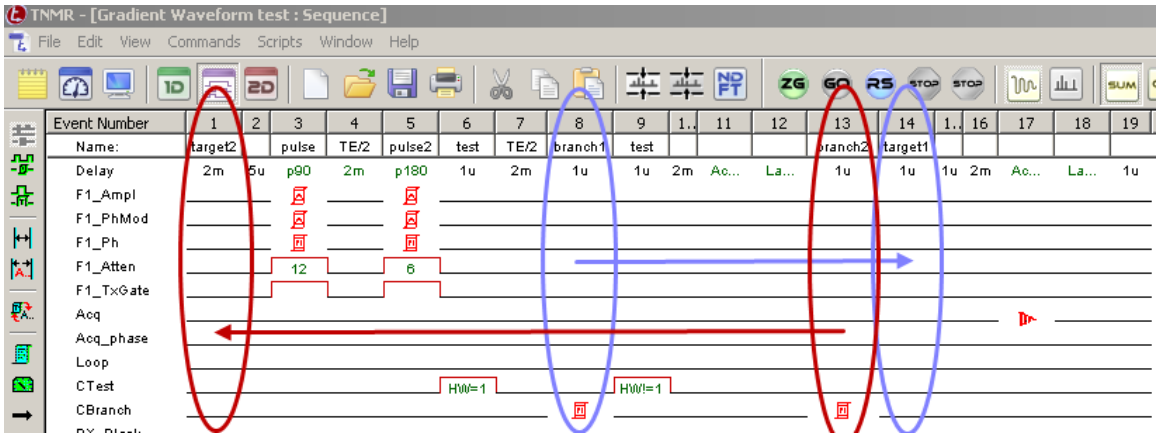
Pulse Programming notes:

There are two condition tests, in events 6, and 9, with opposite logic. The test in event 6 is true if hardware bit 0 is high (logic “1”), while the test in event 9 is false.

The CBranch table in event 8 has one entry, “14”. The table in event 13, likewise has one entry, “1”.

Be sure to count the events executed along each path and set the durations equal in order to maintain constant timing of the sequence.

(Note that this example assumes that the external condition bit is constant for the duration of the scan. This can be achieved, for example, by using an LP output in event 18 to latch in a new value).



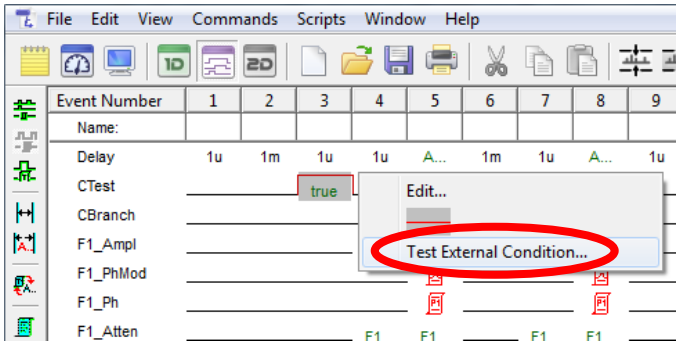
Hardware Interface for Conditional Branching:

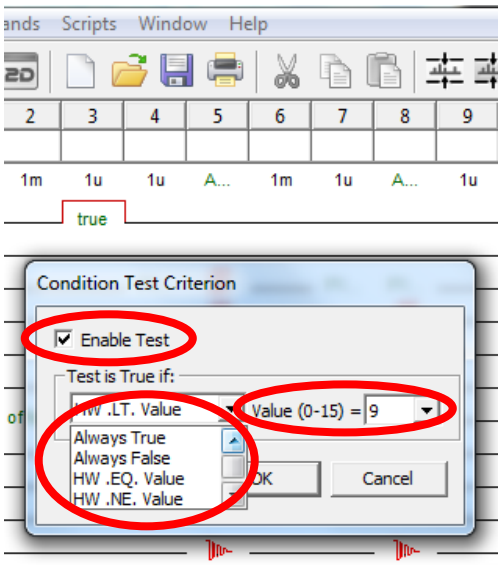
A DB-25 connector labeled “EXT IN” is provided on the Redstone back panel to interface external hardware signals for conditional branch control. Currently four bits are tested allowing 16 values (“conditions”).

EXT IN	
Bit	**Pin**
0	1
1	2
2	3
3	4
GND	25

CTest Logic Conditions:

The TNMR Sequence Editor provides a number of logic options for conditional branch testing.



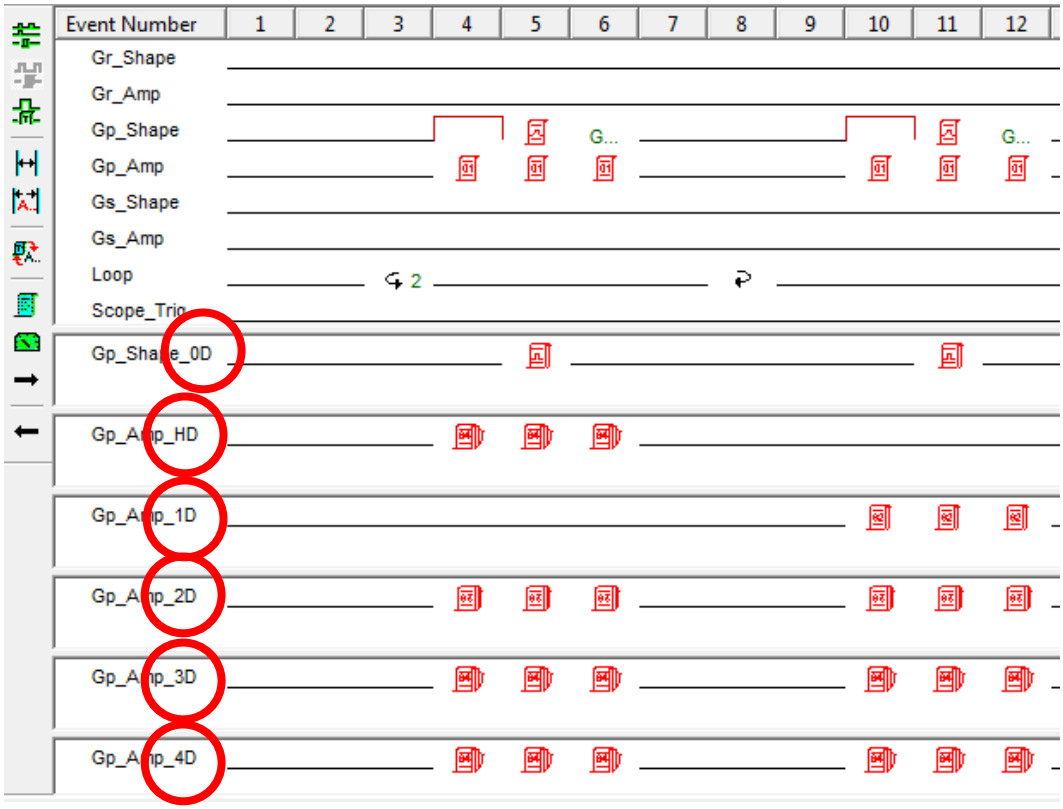


“Test is True if:”	**Expression**
Always True	True for all Values
Always False	False for all Values
HW .EQ. Value	=
HW .NE. Value	≠
HW .GT. Value	>
HW .LT. Value	<
HW .GE. Value	≥
HW .LE. Value	≤

Simultaneous Tables

New functionality has been included in the Redstone architecture that allows for simultaneous waveform (i.e. 0D or shape tables) and multi-dimensional (i.e. 1D, 2D, 3D, 4D, etc) tables. This now allows for example 0, 90, 180, and 270 phase manipulation of a phase-modulated shaped RF pulse by creating phase cycling tables and waveform tables.

In order to support the changes, the graphical pulse sequence editor in TNMR has been modified. All tables including waveform tables (0D), half-D loop tables (HD), and multi-dimensional tables (1D – 4D) are now displayed separately in panes toward the bottom of the sequence editor window. A place-holder icon is still displayed in the main pane.



Modules

Digital Attenuator

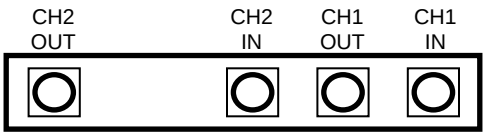
Overview:
The digital attenuator incorporates a two-channel digital attenuator module with two Hittite Microwave Corp. HMC470LP3 digital attenuators. Five-bit control is provided via the cluster backplane. The software interface is accomplished using the Ext_Gain line of the TNMR pulse sequence editor.

The attenuator allows 1.0 dB incremental attenuation from 0 dB to 31 dB (5-bits). However, due to the logic of the device, software control is presented as a "Gain" since a value of 31 (all 5 bits high logic) delivered to the device corresponds to 0 dB attenuation while a value of 0 (all 5 bits low) results in maximum attenuation. NOTE: The two channels are controlled together using the same sequence editor control.

The attenuator hardware is factory installed as an internal component of the Redstone. It may be installed in the System or LO Synthesizer Cluster; it is inserted in the cable path between the RX input on the Redstone back panel and the RX analog receiver input. All cabling has been installed in the factory; no additional installation is required. The SMA Port Assignments are illustrated below.

The /tnmr/config/config.con file should be modified to include a 5-bit control line (e.g. Ext_Gain) for pulse sequence control.

Operating Instructions:
1. Add a "Gain" value or variable on the Ext_Gain line during every acquisition event in the TNMR pulse sequences.
2. For maximum signal, enter a value of 31. For minimum signal, use a variable set to zero. For 10 dB attenuation, enter 31 – 10 = 21.



Two Channel Digital Attenuator
SMA Port Assignments

Utility

Overview:
The 2012 Utility module replaces the previous External Trigger / Rotor Synchronization Module (RSM). It incorporates all of the features of the Ext. Trig / RSM module and is drop-in compatible with it, however, it also adds a number of features including

- Two six-channel analog-to-digital converters (ADCs)
- An on-board temperature sensor
- Conditional branching inputs
- A tachometer

ADCs:
Some of the ADC channels are used for monitoring internal Redstone voltages including ± 5V and ±15V. The analog-to-digital values can be retrieved using the Application-level OLE command *GetAuxDataChannelFormatted(n)* where n is the channel number. For example, the on-board temperature may be monitored by retrieving channel n=2. Two analog signal inputs are provided on the EXT IN DB25 rear-panel connector.

Aux Channel	Signal
0	ADC 0
1	ADC 1
2	Temperature
8	+15V
9	+5V
10	-5V
11	-15V

"EXT IN" DB25 pin	Signal
1 - 8	Conditional Branch bit 0-7
9	ADC 0 input
10	ADC 1 input
11	External trigger TTL
12	Tachometer TTL (RSM)
13	+5V
14 - 25	GND

Triggering and Tachometer:
The external trigger / RSM and tachometer are configurable allowing the user to trigger the Ext Trig pulse-sequence line using either the external trigger TTL input or the Tachometer TTL (RSM) input. Furthermore, the on-board tachometer can monitor either TTL input. For rotors / spinners with multiple tach markings, a Tach divider can also be configured for proper spin rate determination.
The tachometer signal can be retrieved using the Application-level *GetSpinRate()* command.

Tach Mode*	Ext Trig Source	Tachometer Source
0x00n0	ext trig TTL	ext trig TTL
0x00n1	RSM TTL	ext trig TTL
0x00n2	ext trig TTL	RSM TTL
0x00n3	RSM TTL	RSM TTL
**n = Tach divider (hex: 1-F)*		

Apollo.ini Tachometer Configuration

```
[Tach]
ModuleType = Tachometer
; Mode - bits 6..4 - Tach divider (does not apply to ext trig input)
;        bit     1 - Tachometer readout source (1 = tach, 0 = ext trig)
;        bit     0 - Trigger source (1 = tach, 0 = ext trig)
Mode       = 0x0013   ;divider = 1, RSM TTL for tach and trigger
CountInterval = 1.0    ; seconds
LogLevel   = 1
```

LapNMR

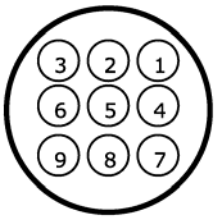
Voltages

Input Power Requirements

Input Voltage:	Portable Chassis AC-DC Adapter:	100 – 240V_{AC}, 50-60 Hz, auto-switching
	2U Chassis w/o Gradient:	100 – 240V_{AC}, 50-60 Hz, auto-switching
	2U Chassis with Gradient:	100 – 120VAC, 200 – 240 VAC, selectable‡
Power:	Portable Chassis:	66 Watts
	2U Chassis w/o Gradient:	125 Watts
	2U Chassis with Gradient:	150 Watts

2U Chassis Auxiliary Voltage Output

AUX 1

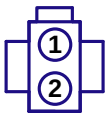


1. +5V
2. 5V GND
3. -5V
4. +15V
5. 15V GND
6. -15V
7-9. N\C

Current: 0.5 Ampere maximum per voltage output

Portable Chassis Auxiliary Voltage Output

AUX OUT

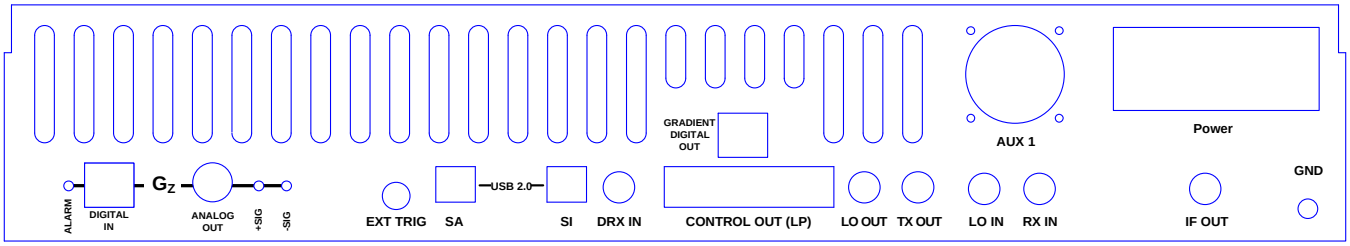


1. **+12 V_{DC}**
2. **12V_{DC} GND**

Current: 0.5 Ampere maximum fused output

‡ Removing the 2U chassis top cover reveals the AC voltage selection switch.

Back Panel Layout—2U Chassis



RX IN:	Receiver input
TX OUT:	Transmitter output
CONTROL OUT (LP):	Pulse programmer-controlled TTL outputs for amplifier blanking and scope triggering; additional lines may be added as required
SA and SI	USB 2.0 connections to computer
EXT TRIG:	External Trigger input; TTL; for external triggering of the LapNMR pulse programmer
G_Z ANALOG OUT:	Analog (Twin-axial BNC) gradient signal output (for systems equipped with single channel gradient)
AUX 1:	+15 V_{DC} output for auxiliary devices including preamplifier

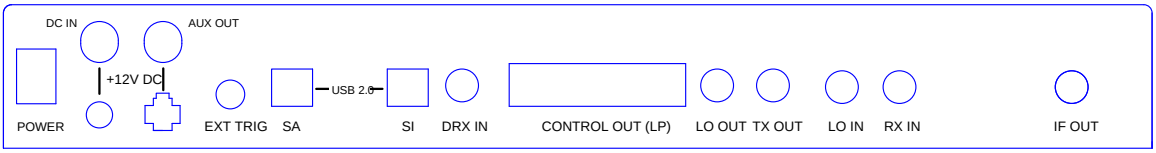
Interconnections

LO OUT to *LO IN*

IF OUT to *DRX IN*

GRADIENT DIGITAL OUT to *DIGITAL IN*
(for systems equipped with single channel gradient)

Back Panel Layout—Portable



RX IN:	Receiver input
TX OUT:	Transmitter output
CONTROL OUT (LP):	Pulse programmer-controlled TTL outputs for amplifier blanking and scope triggering; additional lines may be added as required
SA and SI	USB 2.0 connections to computer
EXT TRIG:	External Trigger input; TTL; for external triggering of the LapNMR pulse programmer
DC IN	$+12V_{DC}$ input
AUX OUT:	$+12\ V_{DC}$ output for auxiliary devices including preamplifier

Interconnections

LO OUT to *LO IN*

IF OUT to *DRX IN*

Specifications

Pulse Programmer

Specification	**Value**
Time resolution (minimum increment)	20 ns
Minimum event duration	160 ns
Maximum event duration	85.8993459 s
Maximum events in sequence	1024
Max depth of nested loops	5
Maximum loop counter	16383
External trigger signal	TTL, any width

Data Acquisition

Specification	**Value**	**Notes**
No. Points 1D	1	Single-Point Acquisition
	even, ≥ 8, and: ≤ host memory	Host memory
	≥ 48 (fast mode)□□ ≤ 4096 (fast mode)	DSP buffer memory on DRX board
No. Points 2D, 3D, 4D	≤ 64k-1	
Max total points per expt.	Depends on host physical memory (max is < 50% of physical memory).	Specify in INI file: example: 192 Mbytes = 24M complex pts for 512M host memory (192Mbyte = 512x512x96)
No. Scans 1D	< 64k	
Continuous scan	No	
Dummy Scans	Yes	Always use ≥ 2 dummy scans
Real-time display	Yes	
Soft “Yellow” stop/restart	No	
Minimum dwell time	60 ns	
Maximum dwell time	21 ms	
Minimum Recycle Delay	See digital receiver section	
Single-point acq	≥ 1 μs / point	

Transmitter (Analog)

Typical measurements with 6dB attenuation at 42.57 MHz (1 Tesla).

Specification	**Value**	**Notes**
Output power	1 Vpp (+4 dBm +/– 3 dB)	Attenuator = 6 dB
Output impedance	50 ohms	
RF Bandwidth		
(– 3 dB)	40 kHz – 120 MHz	
(– 6 dB)	25 kHz – 125 MHz	
Rise time (10-90%)	< 20 ns	

Transmitter (Digital)

Specification	**Value**	**Notes**
Phase control (real-time)	0°, 90°, 180°, 270°	From pulse sequence
Phase change latency (delay)	84 ns	pipeline delay = 28*2.5ns = 70 ns, worst-case t_{pd} = 19.8 ns (U604)
Amplitude control	0.0 – 60.0 dB attenuation	Preset in Apollo.INI file
Amplitude control (real-time)	Select either of two preset values	From pulse sequence (preset in Apollo.INI file)
Amplitude change latency (delay)	1.39 μs	Serial transmission delays
Attenuator phase shift	< 0.2°	30 dB step
	< 2.0°	40 dB step
Frequency control	To ~0.1 Hz	From dashboard
Frequency control (real-time)	Select either of two preset values	From pulse sequence (preset in Apollo.INI file)
Frequency change latency (delay)	2.36 μs	Serial transmission delays
Phase reset latency (delay)	4.66 μs	Serial transmission delays. No RF output for ~2.3 μs during this time

NOTES:

Phase reset, frequency and amplitude changes cannot occur at the same time. All of these actions begin at the rising edge of the relevant LP line pulse and continue for some time. Frequency and amplitude changes also occur on the falling edge of the LP pulse. A delay at least equal to the latency shown above should be included before commencing the next action. Serial transmission times (see above) must be observed in programming these events.

Digital Receiver

All measurements at 12.5 MHz in Normal Mode unless otherwise noted.

Specification	**Value**	**Notes**
Input impedance (nominal)	50 ohms	
Full-scale input signal range	1 Vpp FS (+/-500mV)	
ADC sample rate	50 MHz	
Input frequency range	DC – 180 MHz (+/- 3dB)	
"DC offset" (IF carrier leakage)	< 0.25 bit per scan < 4 μV < -100dBfs	At SW +/- = 50000 Hz and 12.5 MHz IF.
Quadrature ghost	< -70dB	At SW +/- = 50000 Hz and full-scale signal
Phase transient	< 0.02 deg over 100 μsec	
Passband variation	+/- 1% over 70% of SW	Dwell time > 2 μsec, Standard filter
	+/- 1% over 50% of SW	Dwell time > 2.5 μsec, Minimum delay filter
Minimum dwell time	60 ns	
Maximum dwell time	21 ms	
Minimum Recycle Delay	≤ 30 μsec + 16 dwell + 1 μsec/pt	Fast mode
	≤ 31 μsec + 1 dwell	Normal mode
	~ 1 dwell	Serial mode
	~ 1 dwell	Serial mode w/ DSP option

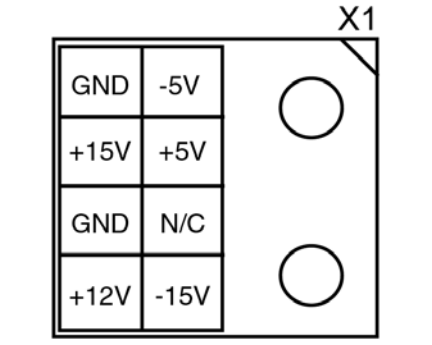
RF Receiver

Specification	**Value**
Input impedance (nominal)	50 ohms
Return Loss	< -30 dB (at 62.5 MHz) < -20 dB 200 kHz – 100 MHz
Max input signal	~ 100mVpp (-15 dBm)
Input frequency range	< 100 kHz – > 120 MHz
Gain control range	> 70 dB
Max gain	~ 80 dB

Power Supply

Specifications apply to 2U LapNMR with Sunpower SPS-100PQ2 switching power supply.

Specification	**Value**	**Notes**
Auxiliary +15V	2.0 A	Available amperage for auxiliary devices (e.g. preamps)
Required currents (if Sunpower supply not used):		
+15 V	2.0 A	Includes +12 V requirement (R5 or an external jumper must be installed)
+5 V	1.5 A	
-5 V	0.2 A	
-15 V	0.0 A	Provided for daughter boards only



8 pin Molex power connector

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